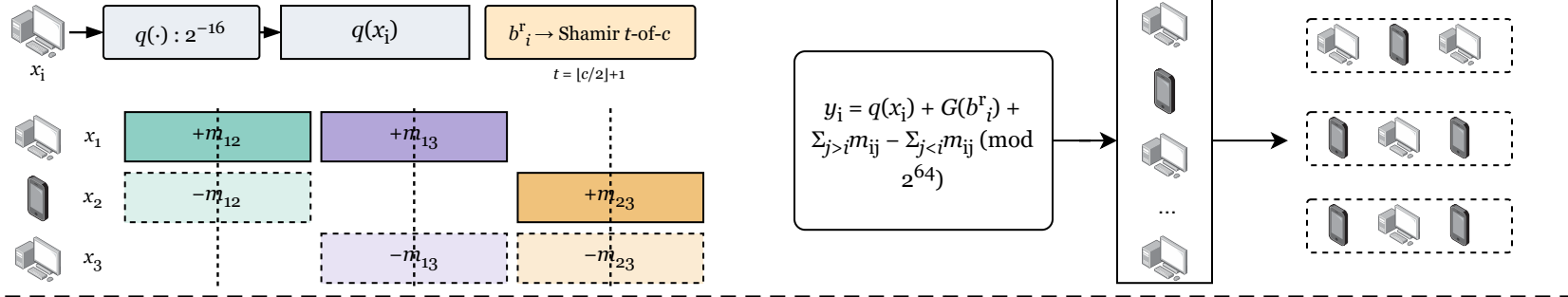


(b) Client masking



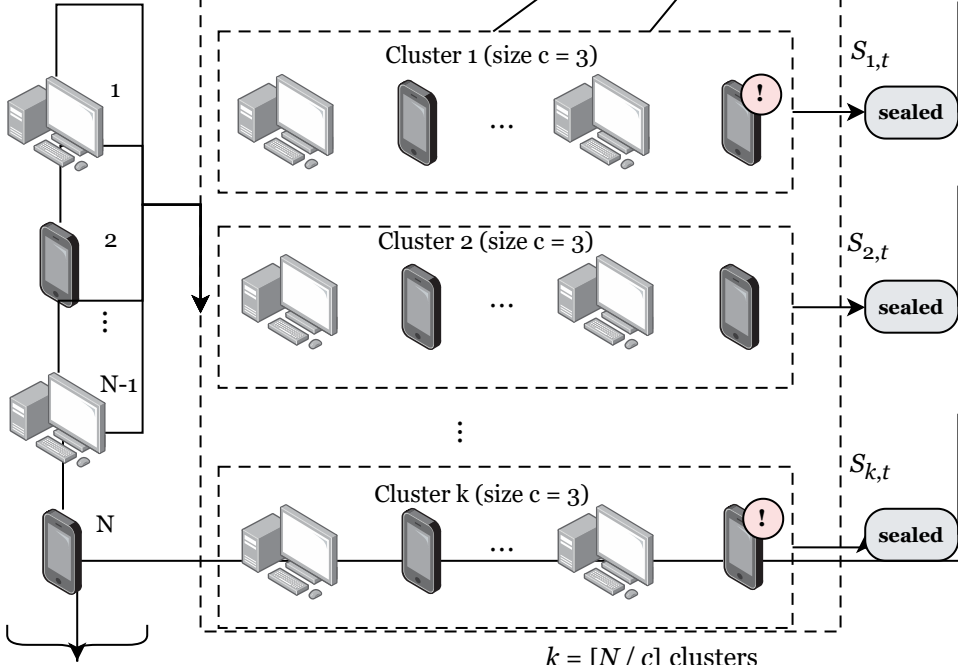
Legends

- ! malicious client
- clean / kept
- flagged / excluded
- sealed masked cluster sum $\rightarrow S_{j,t}$
- privacy boundary (server sees only aggregates)
- C client's group in a round (re-randomized each round)

(c) Secure Aggregation Within Clusters

dropout-tolerant: any $c-t$ may drop
 survivor: t shares
 $\times$ dropout: recover via $H(ss)$

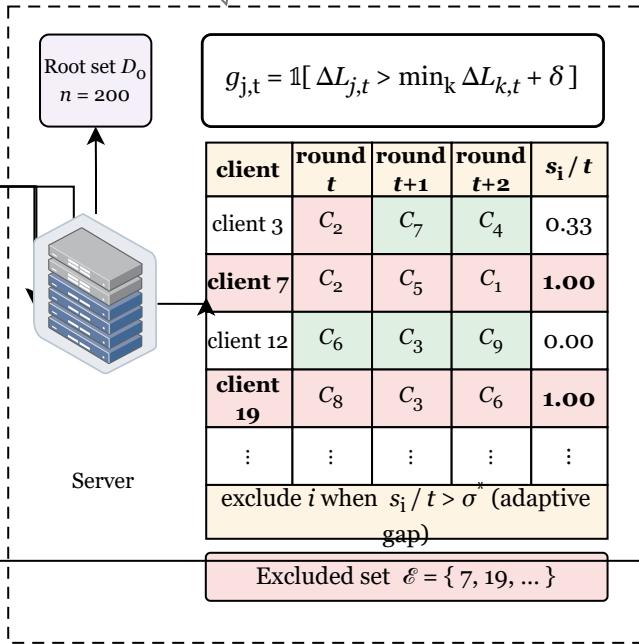
Clients



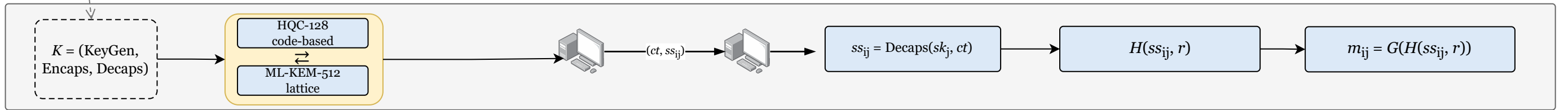
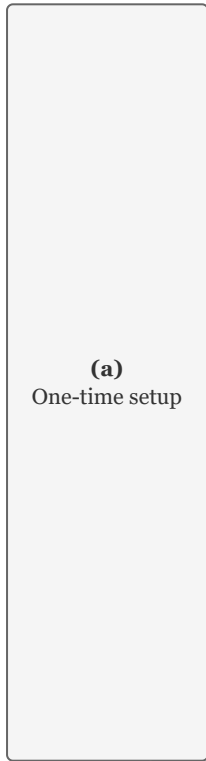
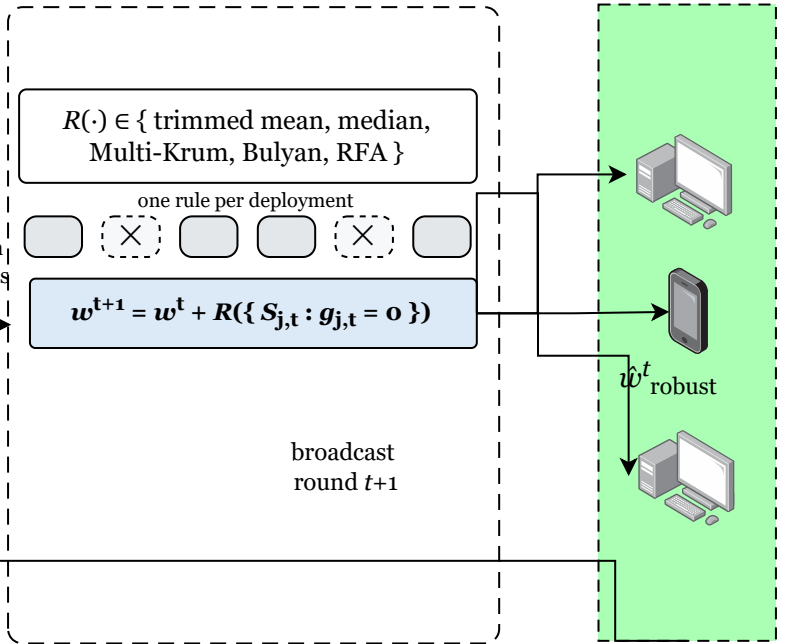
$k = \lceil N/c \rceil$ clusters
 $\Pr[S_j \text{ clean}] = (1-f)^c$

(d) Loss-Probe Flagging & Temporal Group Testing

round t C₂ C₅ C₇ re-randomized each round · client in 1 group/round, overlap over o rounds · no per-round KEM cost
 round $t+1$ C₁ C₃ C₆



(e) Robust Aggregation & Model Update



H : HKDF-SHA256 · G : ChaCha20 · KEMs: HQC / ML-KEM · 15.2 / 3.8 KiB, once